Question 1: Diagnose Underfitting and Overfitting A telecom company develops Decision Tree classifiers to predict customer churn. The following results are obtained:

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# 1..Identify which models demonstrate underfitting, good fit, and overfitting, with justificatic
Underfitting: Training and  validation both accuracy  are low . model are not getting pattern anc
good fit: Training and validation both accuracy are good and  gaps are minimum them both.
overfitting: training accuracy are very high but validation accyracy are low as compare to traini
```

```
# 2..Calculate the training-validation accuracy gap for each model.
FORMULA:->
Gap = Training\ Accuracy - Validation\ Accuracy
gap=68-66%
for each model
 A=2%
 B=3%
 C=22%
 D=31%
```

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# 3..Recommend the most suitable model and justify your decision using generalization performance
i am personally recommended which model is B
Reasons:
1. Training accuracy = 87%.
2. Validation accuracy = 84%.
3. Accuracy gap = only 3%.
4. Model training data ke saath unseen validation data par bhi achha perform kar raha hai.
5. Isliye Model B ki generalization performance achhi hai.
```

Question 2: Advanced Model Selection and Inference Pipeline

An e-commerce company wants to predict whether a customer will purchase a product.

Its dataset contains 15,000 records, and the team compares models with different hyperparameters.

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# Diagnose the likely causes of the performance drop using overfitting and the bias–variance trac
Overfitting :- Model training data ko memorize kar raha hai.
Bias:-Variance Tradeoff : Model mein High Variance hai.
Performance Drop : Accuracy 99% → 81% → 65% hone ke reasons.
Conclusion : Model overfitting aur poor generalization se suffer kar raha hai.
```

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# Design a complete ML workflow covering data splitting, preprocessing, model training, hyperparam
1. Data Splitting :- Dataset ko Training (70%), Validation (15%) aur Testing (15%) mein divide kar

2. Data Preprocessing :- Missing values handle karna, duplicate records remove karna aur categoric

3. Model Training :- Training data par suitable ML algorithm, jaise Decision Tree Classifier, trai

4. Hyperparameter Tuning :- Different hyperparameters (max_depth, min_samples_split) test karke be

5. Model Validation :- Validation data par model ki performance check karna using Accuracy, Precis

6. Best Model Selection :- Different models compare karke best-performing model select karna aur o

7. Final Testing :- Unseen test data par selected model ki final performance evaluate karna.
```

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8. Final Inference :- New customer data par prediction karna:
- 1 = Customer will purchase.
- 0 = Customer will not purchase.
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